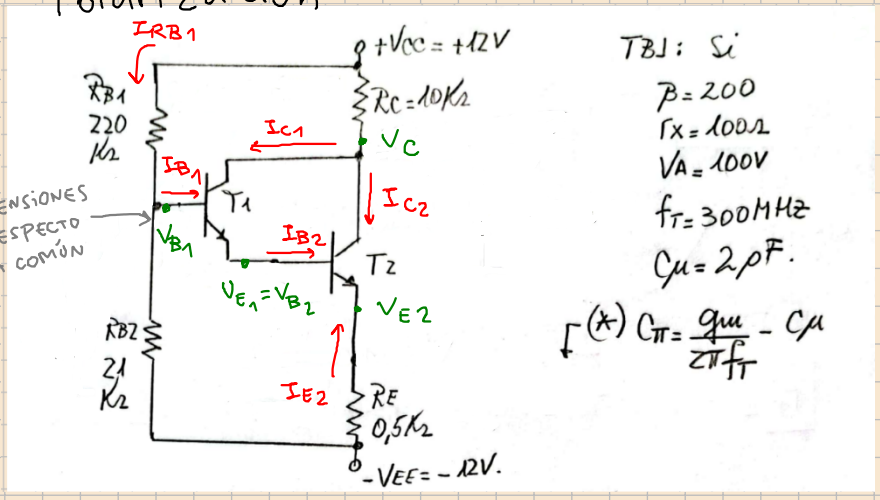
Polarización



TBJ: Si
$\beta = 200$
$r_x = 100\Omega$
$V_A = 100V$
$f_T = 300MHz$
$C_\mu = 2pF.$
(*) $C_\pi = \frac{g_m}{2\pi f_T} - C_\mu$

$$V_{BB} = \frac{2V_{CC} \cdot R_{B2}}{R_{B1} + R_{B2}} - V_{CC}$$

$$V_{BB} = \frac{24V \cdot 21k\Omega}{220k\Omega + 21k\Omega} - 12V$$

$$V_{BB} \simeq -9,9V$$

$$I_{RB_1} = \frac{2V_{CC}}{R_{B1} + R_{B2}} = \frac{2 \cdot 12V}{220k\Omega + 21k\Omega} = 99,6\,\mu A$$

Suponiendo $I_{B1} << I_{RB1} \Rightarrow V_{B1} \approx V_{BB} = -9,9V$

Suponiendo ambos TBJ en MAD: $V_{BE(ON)} = 0,7V$

$$\Rightarrow V_{E_1} = V_{B2} = V_B - V_{BE(ON)} = -9,9V - 0,7V \approx -10,61V \approx -10,6V$$

$$V_{E2} = V_{B2} - V_{BE(ON)} = -10,61V - 0,7V \simeq -11,31V \approx -11,3V$$

$$I_{E2} = \frac{-V_{CC} - V_{E2}}{R_E} = \frac{-12V + 11,3V}{500\Omega} \simeq -1,38\,mA \approx -1,4\,mA$$

aproximo $I_{E_2} \approx -I_{C_2} \Rightarrow I_{C_2} \approx 1,4mA \Rightarrow I_{B2} \approx \frac{I_{C2}}{\beta} =$

$$\Rightarrow I_{B2} \approx \frac{I_{C2}}{\beta} = \frac{1,4mA}{200} = 6,91\,\mu A = I_{C1}$$

$$I_{B_1} = \frac{I_{C1}}{\beta} = \frac{6,91\mu A}{200} \approx 34,56\,nA$$

→ NO SÉ SI SE PRENDE ESTE TBJ 😵

Se ve que $I_{B1} << I_{RB1} \Rightarrow V_{B1} \approx V_{BB}$ ✓

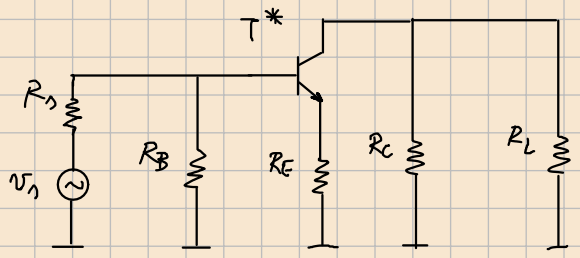
$$V_C = V_{CC} - (I_{C_1} + I_{C_2}) R_C \approx V_{CC} - I_{C_1} R_C = 12V - 1,4mA \cdot 10k\Omega = -1,826V$$

$$V_{CE_1} = V_C - V_{E_1} = -1,83V + 10,6V = 8,77V > V_{CE\,SAT}$$ ✓

$$V_{CE_2} = V_C - V_{E_2} = -1,83V + 11,3V = 9,47V > V_{CE\,SAT}$$ ✓

AMBOS TBJ en MAD ✓

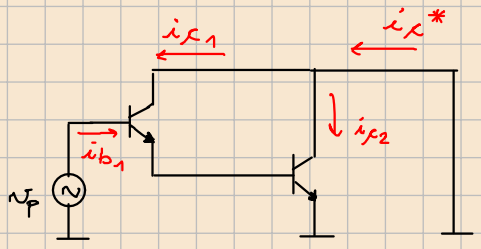
En señal:



con $R_B = R_{B1} // R_{B2} = 220k\Omega // 21k\Omega = 19{,}2k\Omega$

$R_{ca} = R_C // R_L = 10k\Omega // 10k\Omega = 5k\Omega$

Halla T^*



- $g_m^* = \left.\dfrac{i_c^*}{v_{be}^*}\right|_{v_{ce}^*=0}$

$$g_m^* = \frac{i_{c_1} + i_{c_2}}{i_{b_1}(r_{\pi_1} + \beta r_{\pi_2})} = \frac{i_{c_2}\left(1 + \frac{1}{\beta}\right)}{\frac{i_{c_2}}{\beta^2}\left(r_{\pi_1} + \beta r_{\pi_2}\right)}$$

$$i_c^* = i_{c_1} + i_{c_2} = i_{c_2}\left(1 + \frac{1}{\beta}\right)$$

$$i_{b_1} = \frac{i_{c_1}}{\beta} = \frac{i_{c_2}}{\beta^2}$$

$$g_m^* = \frac{1 + 1/\beta}{\frac{1}{\beta^2}\left(\frac{\beta^2}{g_{m_2}} + \frac{\beta^2}{g_{m_2}}\right)}$$

$$r_{\pi_1} = \frac{\beta}{g_{m_1}} = \frac{\beta}{\frac{g_{m_2}}{\beta}} = \frac{\beta^2}{g_{m_2}} = \beta r_{\pi_2}$$

$$r_{\pi_2} = \frac{\beta}{g_{m_2}}$$

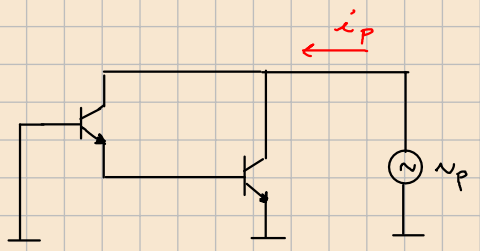
$$g_m^* \approx \frac{1}{\frac{2}{g_{m_2}}} = \frac{g_{m_2}}{2}$$

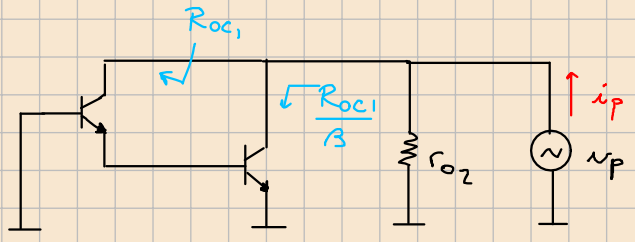
También: $g_m^* = \dfrac{i_c^*}{v_{be}^*} \approx \dfrac{i_{c_2}}{2 v_{be_2}} = \dfrac{g_{m_2}}{2}$

- $\beta^* = \dfrac{i_c^*}{i_b^*} \approx \dfrac{i_{c_2}}{i_{b_1}} = \dfrac{i_{c_2}}{\frac{i_{c_2}}{\beta^2}} = \beta^2$

- $r_\pi^* = \overbrace{r_{\pi_1} + \beta r_{\pi_2}}^{r_{\pi_1}} = r_{\pi_1} + r_{\pi_1} = 2 r_{\pi_1}$

- $r_o^* = \left.\frac{v_p^*}{i_p^*}\right|_{v_i = 0}$





$$r_o^* = r_{o2} /\!/ R_{oc1} /\!/ \frac{R_{oc1}}{\beta}$$

$$r_o^* \approx r_{o2} /\!/ \frac{R_{oc1}}{\beta}$$

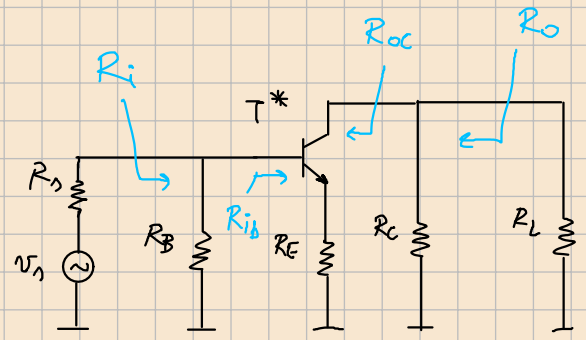
$$r_o^* \approx r_{o2} /\!/ \frac{r_{o1}\left(1 + g_{m1} r_{\pi 2}\right)}{\beta} = r_{o2} /\!/ r_{o2}\left(1 + g_{m1} \cdot \frac{r_{\pi 1}}{\beta}\right)$$

$$r_{o1} = \frac{V_A}{I_{C1}} = \frac{\beta V_A}{I_{C2}} = \beta r_{o2} \Rightarrow \frac{r_{o1}}{\beta} = r_{o2}$$

$$r_o^* \approx r_{o2} /\!/ r_{o2}\left(1 + \frac{1}{r_{\pi 1}} \cdot r_{\pi 1}\right) = r_{o2} /\!/ 2 r_{o2} = \frac{2 r_{o2}}{3}$$

T	I_{CQ} [A]	g_m [S]	r_d [Ω]	r_x [Ω]	r_π [Ω]	r_o [Ω]	C_μ [pF]	C_π [pF]
T_1	6,91µ	267µ	3k74	100	748k	14M5	2	-1,86
T_2	1,38m	53,5m	18,7	100	3k74	72k3	2	26,4
T^*	1,38m	26,7m	37,4	100	1M50	48k2	2	12,2

Ahora sí, parámetros de señal:

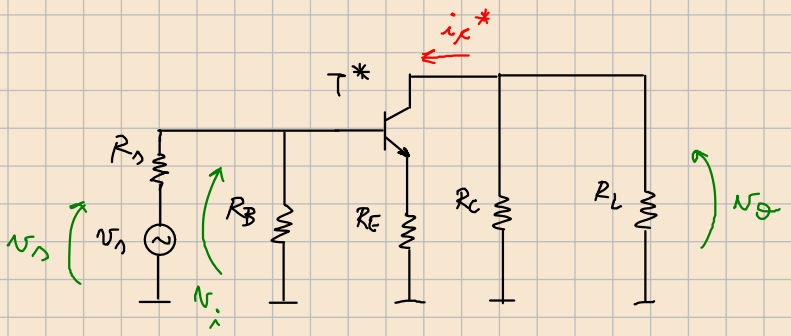


$$R_{ib} = r_\pi^* + \beta^* R_E = 6{,}50\ M\Omega$$

$$R_i = R_B \,//\, R_{ib} \approx R_B = 19{,}2\ k\Omega$$

$$R_{oc} \approx r_o^* \left(1 + g_m^* R_E\right) = 48k2\,\Omega \left(1 + 26{,}7\ mS \cdot 500\,\Omega\right) = 691{,}7\ k\Omega$$

$$R_o = R_{oc} \,//\, R_C \approx R_C = 10\ k\Omega$$

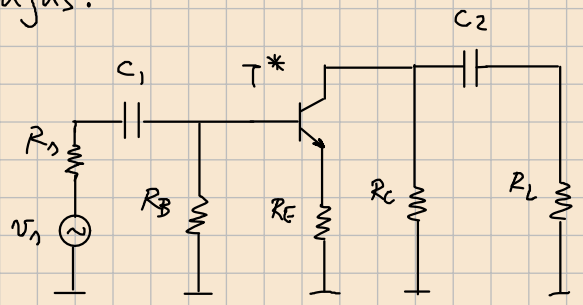


$$A_v = \frac{-i_c^*}{v_i} = \frac{-g_m^* r_{be}^* R_{ca}}{r_{be}^* + g_m^* r_{be}^* R_E} = \frac{-g_m^* R_{ca}}{1 + g_m^* R_E} = \frac{-26{,}7mS \cdot 5k\Omega}{1 + 26{,}7mS \cdot 500\Omega} = -9{,}3$$

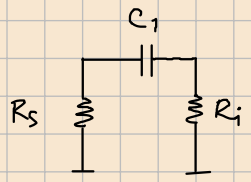
$$A_{v_S} = \frac{v_o}{v_S} = \frac{v_o}{v_i}\,\frac{v_i}{v_S} = A_v \cdot \frac{1}{v_S} \cdot \frac{v_S \cdot R_i}{R_i + R_S} = -5 \cdot \frac{19k2\,\Omega}{19k2\,\Omega + 5k\Omega} = -7{,}4$$

frecuencias bajas:

- f_L



C_1:

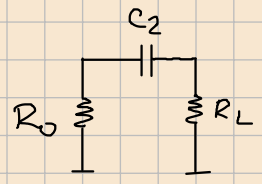


$$\tau_1 = C_1 \cdot (R_S + R_i) = 100\,nF\,(5\,k\Omega + 19{,}2\,k\Omega)$$

$$\tau_1 = 2{,}42\,ms$$

$$f_{L1} = \frac{1}{2\pi\tau_1} = 66\,Hz$$

C_2:



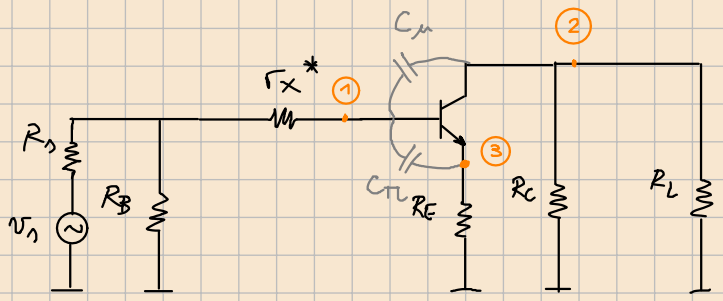
$$\tau_2 = C_2\,(R_L + R_O) = 100\,nF\,(10\,k\Omega + 10\,k\Omega)$$

$$\tau_2 = 2\,ms$$

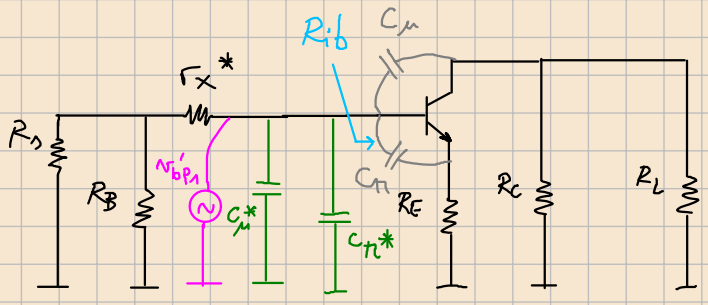
$$f_{L2} = \frac{1}{2\pi\tau_2} = 80\,Hz$$

$$\Rightarrow \quad f_L = f_{L1} + f_{L2} = 66\,Hz + 80\,Hz = 146\,Hz$$

frecuencias altas con T^* :



- NODO 1:



$$C_\mu^* = C_\mu \left(1 - \frac{v_e}{v_{bp_1}'}\right)$$

$$\frac{v_o}{v_{bp_1}'} = \frac{-g_m^* v_{be} R_{ca}}{v_{be} + g_m^* v_{be} R_E}$$

$$\frac{v_o}{v_{bp_1}'} = -9,3$$

$$\Rightarrow C_\mu^* = 2pF\left(1 + 9,3\right) = 20,6\ pF$$

$$C_\pi^* = C_\pi \left(1 - \frac{v_e}{v_{bp_1}'}\right) \quad , \quad \frac{v_e}{v_{bp_1}'} = \frac{g_m^* v_{be} R_E}{g_m^* v_{be} R_E + v_{be}} = \frac{g_m^* R_E}{1 + g_m^* R_E}$$

$$C_\pi^* = 12,2pF\left(1 - 0,93\right)$$

$$\frac{v_e}{v_{bp_1}'} = 0,93$$

$$C_\pi^* = 0,85\ pF$$

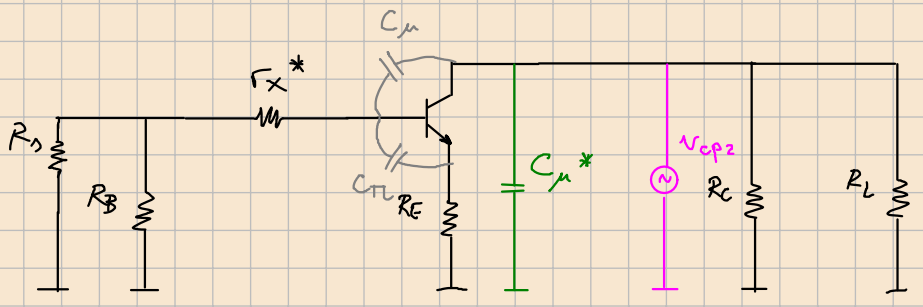
$$C_1 = C_\pi^* + C_\mu^* = 20,6\ pF + 0,85\ pF = 21,5\ pF$$

$$R_{Th_1} = \overbrace{\left[r_x + \left(R_1 /\!/ R_B\right)\right]}^{R} /\!/ R_{ib} \quad \overset{R_{ib} \gg R}{\approx} \quad r_x + R_1 /\!/ R_B = 100\Omega + 5k\Omega /\!/ 19,2k\Omega$$

$$R_{Th_1} = 4k1\Omega$$

$$\tau_1 = C_1 R_{Th_1} = 87,4ns$$

• Nodo 2



$$C_\mu^* = C_\mu \left(1 - \frac{v_{cp2}}{v_b}\right)$$

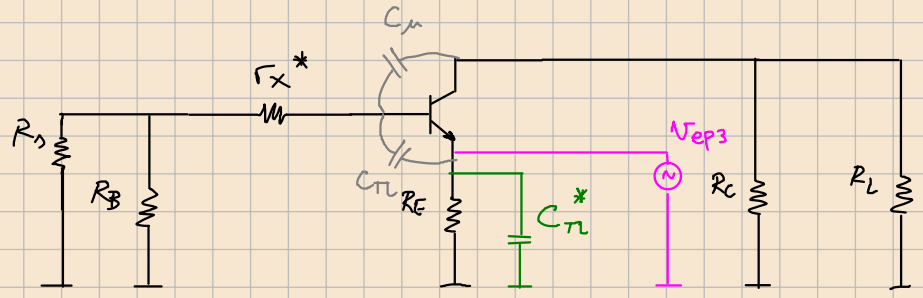
$$\frac{v_{cp2}}{v_b} \longrightarrow 0$$

$$C_\mu^* \simeq C_\mu = 2pF$$

$$R_{Th2} = R_C // R_L // r_o^* = 5k\Omega // 48k2\Omega = 4k5\Omega$$

$$\tau_2 = C_\mu^* R_{Th2} = 9\,ns$$

• Nodo 3:



$$C_\pi^* = C_\pi \left(1 - \frac{v_b}{v_{ep3}}\right)$$

$$\frac{v_b}{v_{ep3}} = \frac{-i_b \left(r_x + R_1 // R_B\right)}{i_C R_E}$$

$$\frac{v_b}{v_{ep3}} = \frac{r_x + R_1 // R_B}{\beta R_E}$$

$$\frac{v_b}{v_{ep3}} = \frac{4k1\Omega}{200 \cdot 500\Omega} = 41\,m$$

$$C_\pi^* = 12,2pF \left(1 - 41\,m\right) = 11,7\,pF$$

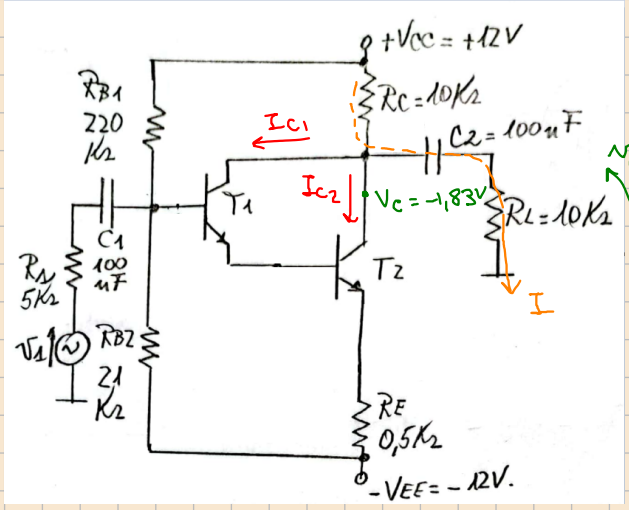
$$R_{Th3} = \frac{r_\pi^* + r_x^* + R_1 // R_B}{\beta} // R_E$$

$$R_{Th3} \simeq \frac{r_\pi^*}{\beta} // R_E = 7k5\Omega // 500\Omega \simeq 467\Omega$$

$$\tau_3 = C_\pi^* R_{Th3} = 5,5\,ns$$

$$f_H = \frac{1}{2\pi\left(\tau_1 + \tau_2 + \tau_3\right)} = \frac{1}{2\pi\left(87,4\,ns + 9\,ns + 5,5\,ns\right)} = 1,57\ MHz$$

$\hat{v}_o$ MÁX sin recorte:



$v_o = v_o + v_o^{=0} = v_o$

Suponiendo que T_2 limita

- T_2 va al corte $\Rightarrow T_1$ corta

$I_{C_2} = I_{C_1} = 0\ A$

$V_{CC} - I R_C - V_C - I R_L = 0$

$I = \frac{V_{CC} - V_C}{R_C + R_L} = \frac{12V + 1,83V}{10k\Omega + 10k\Omega}$

$I = 692\,\mu A$

$\Rightarrow \hat{v}_o^+ = I R_L = 692\,\mu A \cdot 10k\Omega = 6,92\,V$

- T_2 Satura:

Supongo $V_{CE\,SAT\,incipiente} = 0,7V$

$\Rightarrow V_{BC_2} = 0 \Rightarrow V_{CE_1} = 0 \Rightarrow T_1$ Saturadísimo

T_1 es el limitante en saturación

Supongo $V_{CE_1} = 0,7V$ para que no sature

$\Rightarrow V_{CB_2} = 0,7$

$\Rightarrow V_C' = V_{BB} - V_{BE(ON)} + V_{CB} = -9,9V - 0,7V + 0,7V$

$V_C' = -9,9V$

$\Rightarrow \hat{v}_o^- = V_C' - V_C = -9,9\,V + 1,83\,V \cong -8\,V$

$\therefore$ Máxima: $\hat{v}_o = 8V$

Simétrica: $\hat{v}_o = 6,9V$

MÁX $v_i = \frac{\hat{v}_o}{|A_v|} = \frac{8V}{9,3} = 860\,mV$

Simétrica $v_i = \frac{\hat{v}_o}{|A_v|} = \frac{6,9V}{9,3} = 742\,mV$

